

Primitive Element Theorem

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Problem 1. Show that for a field M between K and L , $L=M$ iff any $g : L \rightarrow A$ which is the identity on M is the identity.

Proof. $\implies$: Suppose $L = M$. Then $\forall g : L \rightarrow A$, where $g|_M = id$. Since M is identically L , if $g|_M = id \implies g|_L = id \implies g = id$.

$\impliedby$: Suppose $\forall g \in L \rightarrow A \ni g|_M = id \implies g = id$. Assume the contrary, $L \neq M$. Then $\exists l \in L \setminus M$. Define the ring homomorphism, $h : L \rightarrow A \ni h(l) = 0$ and $h|_M = id_M$. Thus $h|_M = Id$ but $h \neq id$ which is a contradiction. Thus $L = M$. $\square$

Problem 2. Now we prove $L = K[z]$ for some z . Explain why we can assume $L = K[s, t]$.

Proof. Base Case: Suppose $L = K[a]$, we are done.

Inductive Hypothesis: Suppose $\forall a_1, \dots, a_k \in A$ where a_i is separable that $K[a_1, \dots, a_{k-1}] = K[s]$ for some $s \in A$.

Inductive Step: Suppose $L = K[a_1, \dots, a_k]$ where a_i is separable over K . Let $M = K[a_1, \dots, a_{k-1}]$. By the inductive hypothesis, $M = K[s]$ for some $s \in A$. Thus $L = K[s, a_k]$ and we have reduced to the 2 element extension of K . Thus if we prove $L = K[s, a_k] = K[b]$ for $b \in A$, we are done.

Thus it suffices to assume that $L = K[s, t]$ for some $s, t \in A$ separable . $\square$

Problem 4. We consider $s + kt$ for k in K , and we will show that $L = K[s + kt]$ for some k (in fact it will work for all but finitely many k in K). We do it by picking k so that there are no $g : K[s + kt] \rightarrow L$ which are the identity on K .

Explain why $E := \{g : L \rightarrow A \mid g(k) = k \text{ for all } k \in K\}$ is finite. Take $g \in E$, g not the identity. Show there is at most one $k(g)$ so that $g(s + kt) = s + kt$. Now take x in K which is not one of the $k(g)$, for g in E . Why does such an x exist? Explain why $g(s + xt) = s + xt$ iff $g \in E$ is the identity. Conclude that $K[s + xt] = L$, for all but finitely many x .

Proof. $E = \text{Ext}_K^L(id)$, and $|E| = |\text{Ext}_K^L(id)| \leq [L : K]$. Thus $\text{Ext}_K^L(id)$ is finite. $\forall g \in E \ni g \neq id$. Let $k(g) \in K \ni g(s + kt) = s + kt$. Thus $g(s) + kg(t) = s + kt \implies g(s) - s = k(t - g(t))$. Since $L = K[s, t]$, if $g(s) = s$ and $g(t) = t$, $\forall as + bt + c \in K[s, t]$, $g(as + bt + c) = as + bt + c$ and thus $g = id$. Thus g is not the identity on both s and t . If $g(t) = t$, then there exists no $k \in K$. Otherwise, $k = \frac{g(s) - s}{t - g(t)}$ which is either in K and is unique or is not in K and thus invalid. Thus there is either 0 or one $k(g)$ for any g where g is not the identity. Since E is finite and for each $g \in E$ where $g \neq id$, there is at most 1 $k(g)$ associated with it, the $\{k(g) \mid \forall g \in E\}$ is finite.

Suppose, $\exists g \in E \ni g(s + xt) = s + xt$. Assume the contrary, $g \neq id$, then $x = k(g)$ and $x \in \{k(g) | \forall g \in E \setminus \{id\}\}$ since $g \neq Id$ which is a contradiction. Thus $g = id$.

Suppose g is the identity, then $g(s + xt) = s + xt$.

Thus the only function $g : L \rightarrow A$ where $g|_{K[s+xt]} = id$ is the identity function. Thus by problem 1, $L = K[s + xt]$. $\square$

Let L/K as usual be a finite extension. Recall we say an element a of L is separable over K if its min poly has no multiple roots (in any field containing K).

Problem 5. Say $L = K[a_1, \dots, a_r]$. Show if all a_i are separable, then every a in L is separable. In which case we say L is separable over K .

Proof. Base Case: Suppose $\forall a \in A$ where a is separable over K , by the field symmetries homework, $|\text{Ext}_K^{K[a]}(id)| = [K[a] : K]$. By the same homework, that means $\forall l \in L$, l is separable.

Inductive Hypothesis: Suppose $\forall a_1, \dots, a_{k-1} \in A$ where a_i is separable over K for all i , that $K[a_1, \dots, a_{k-1}]$ is separable over K .

Inductive Step: Suppose $\forall a_1, \dots, a_k \in A$ where a_i is separable over K for all i . By the induction hypothesis, $M = K[a_1, \dots, a_{k-1}]$ is separable. By the Field Symmetries homework $|\text{Ext}_K^M(id)| = [M : K]$. Let $r : \text{Ext}_K^{K[a_1, \dots, a_k]}(id) \rightarrow \text{Ext}_K^M(id)$. By the field Symmetries homework, $\forall q \in \text{Ext}_K^M(id)$, $r^{-1}(q) = \text{Ext}_M^{K[a_1, \dots, a_k]}(q)$. Furthermore, $K[a_1, \dots, a_k] = M[a_k]$. Since a_k is separable over K , it is separable over M because the minimal polynomial of a_k over M must divide the minimal polynomial of a_k over K . Thus by field symmetries homework, $\text{Ext}_M^{K[a_1, \dots, a_k]}(q) = [K[a_1, \dots, a_k] : M]$ for all $q \in \text{Ext}_K^M(id)$. Thus every fiber is the same size, thus r is surjective. Thus $\text{Ext}_K^{K[a_1, \dots, a_k]}(id) = \sum_{q \in \text{Ext}_K^M(id)} \text{Ext}_M^{K[a_1, \dots, a_k]}(q) = \sum_{q \in \text{Ext}_K^M(id)} [K[a_1, \dots, a_k] : M] = [M : K][K[a_1, \dots, a_k] : M] = [K[a_1, \dots, a_k] : K]$. By the field symmetries homework, $K[a_1, \dots, a_k]$ is separable.

Thus L is separable. $\square$

Problem 6. Let g be a monic poly in K , with no double roots in any field extension. Let L be a splitting field for K , this means that g has all its roots in L , and L is generated over K by the roots of g (note this means there is no proper subfield of L in which g has all its roots). Let a be any element of L . Show its minimal polynomial has all its roots in L , and no double roots. Why is this surprising?

Proof. Let the roots of g be $a_1, \dots, a_k$. By assumption, $a_1, \dots, a_k$ are distinct. Let $L = K[a_1, \dots, a_k]$. $\forall i \in [1, \dots, k]$, since a_i is a root of g and thus p_{a_i} , the minimal polynomial of a_i , divides g , thus since g has distinct roots, p_{a_i} also has distinct roots, and since all the roots of g are in $K[a_1, \dots, a_k]$, all the roots of p_{a_i} are in $K[a_1, \dots, a_k]$. Thus a_i is separable. By problem 5, $K[a_1, \dots, a_k]$ is separable and thus $\forall l \in K[a_1, \dots, a_k]$, the minimal polynomial of l has all its roots in $K[a_1, \dots, a_k]$ and all the roots are distinct. $\square$